Pedaprolu Lakshmi Kushal

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	5.76
Sri Chaitanya Junior Kalasala, Gudavalli, Vijayawada Rural	Class XII (Board of Intermediate Education - AP)	2024	98.30%
Sri Chaitanya High School, Punadipadu, Krishna District - AP	Class X (Board of Secondary Education - AP)	2022	95.50%

SKILLS

- **Programming Languages:** C, Python, SQL
- **Libraries & Frameworks:** NumPy, Pandas, Matplotlib, PyTorch
- **Version Control:** Git, GitHub
- **Operating Systems:** Linux, Windows
- **Development Tools:** Jupyter Notebook, Google Colab, Visual Studio Code
- **Core Concepts:** Data Structures and Algorithms, Object-Oriented Programming (OOP), Database Management System (DBMS), Machine Learning, Deep Learning
- **Soft Skills:** Communication, Problem Solving, Time Management

RELEVANT COURSES

- **Artificial Intelligence:** Foundations of Machine Learning, Deep Learning, Introduction to Modern AI, AI for Humanity, Programming for Artificial Intelligence
- **Computer Science:** Introduction to Programming, Data Structures and Algorithms, Algorithms, Database Management System (DBMS)
- **Mathematics:** Discrete Mathematics, Calculus, Metric Spaces, Optimization, Probability and Random Variables

PROJECTS

Neural Networks From Scratch

- Developed **Perceptron, ADALINE, Multi-Layer Perceptron (MLP)**, and XOR classification models from scratch using NumPy.
- Implemented **forward propagation, backpropagation**, weight initialization, gradient computation, and **L1/L2 regularization**.
- Developed activation functions including **Sigmoid, Tanh, ReLU, Leaky ReLU, and Softmax** with analytical derivatives.
- Integrated optimizers such as **SGD, Momentum, RMSProp, and Adam** with hyperparameter tuning and training visualization.

Convolutional Neural Network (CNN) From Scratch

- Built a complete **CNN framework** implementing convolution, pooling, activation, flattening, batch normalization, dropout, and fully connected layers.
- Implemented configurable architectures supporting multiple kernel sizes, padding strategies, activation functions, and optimization algorithms.
- Trained and evaluated models on the **CIFAR-10** dataset while comparing optimizer performance and improving model generalization.
- Analyzed learned feature representations through **feature map visualization** and monitored training using loss and accuracy metrics.

Recurrent Neural Network (RNN) From Scratch

- Implemented **Vanilla RNN** and **Long Short-Term Memory (LSTM)** networks using **Backpropagation Through Time (BPTT)**.
- Developed a **Transformer Encoder** with multi-head self-attention, positional encoding, residual connections, layer

normalization, and feed-forward layers.
- Implemented **Linear and Selective State Space Models (SSMs)** for efficient long-context sequence modeling and sentiment classification.
- Applied gradient clipping, trainable embeddings, sequence batching, and evaluated models using training loss, validation accuracy, and ablation studies.

EXTRACURRICULAR

- Actively participated in the **National Service Scheme (NSS)**, contributing to **community service** and various **social welfare initiatives**.
- Trained in **Classical Keyboard Music**, developing **discipline**, **creativity**, and **performance skills** through regular practice.